

Utilizing Attention-Based Multi-Encoder-Decoder Neural Networks for Freeway Traffic Speed Prediction

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Abstract—Speed prediction is a crucial yet complicated task for intelligent transportation systems. The challenge derives from the complex spatiotemporal dependencies of traffic parameters. In the past few years, deep neural networks have achieved the best traffic speed prediction performance. However, most models depend on short-term input sequences to predict short/long-term traffic speed (e.g., predicting speed for the next hour using data from the past hour). These models fail to consider the daily and weekly periodic behavior of traffic. Another problem posed by neural networks is the lack of interpretability as they often operate as “black boxes”. In this paper, an attention-based multi-encoder-decoder (Att-MED) model is proposed to predict traffic speed. The model uses convolutional-LSTMs to capture the spatiotemporal relationship of multiple input sequences, namely short-term, daily and weekly traffic patterns. The model also employs an LSTM to model the output predictions sequentially. Furthermore, attention mechanism is used to weigh the contribution of each traffic sequence towards the output predictions. The proposed network architecture, when trained end-to-end, results in a superior prediction accuracy compared to baseline models. In addition to contributing towards performance, the attention mechanism creates weight values, which when visualized, provide insights into the decision-making process of the neural network, and consequently produce explainable outputs. Att-MED’s extracted attention weights highlight the contribution of daily and weekly periodic input towards speed prediction.

Index Terms—Traffic forecasting, multi-sequence, encoder-decoder, attention, explainable neural networks.

I. INTRODUCTION

TRAFFIC speed prediction is an essential component of Intelligent Transportation Systems (ITS). It contributes towards crucial transportation applications like navigation guidance, traffic scheduling and traffic management [1]. Traffic speed prediction is a difficult problem since it is necessary to consider both the temporal dependency between traffic parameters and the spatial connection between traffic parameters at different parts of the road network. Furthermore, this spatiotemporal relationship is stochastic and highly non-linear in nature, making it difficult to model accurately [2].

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The ubiquity of traffic sensors on the road network has led to an explosion of traffic data, which has in turn fueled a growing body of research in data-driven modeling of the speed prediction problem [3]. These types of models can be classified into two types: classical statistical models and machine intelligence models [4]. Statistical models, specifically parametric models, are favored for their interpretability and relative simplicity. On the other hand, machine intelligence models are non-parametric models which make little to no assumptions regarding the input variables. Hence, they have the ability to model strong generalizations in pattern recognition. Due to the advancement in computation space and time efficiency, Artificial Neural Networks (ANNs), specifically deep learning techniques, have been widely applied in traffic speed prediction during the past few years. They have since demonstrated superior performance compared to classical statistical models and machine learning algorithms.

Despite the recent achievements in deep learning-based traffic speed prediction, there are some major limitations in the literature. Firstly, most research efforts focus on a short-term input sequence, like data from the past minutes/hours, when in fact, traffic speed exhibits cyclic behavior across consecutive days (specifically weekdays). In addition, there’s a periodic correlation between traffic patterns during the same day-of-week across successive weeks [5]–[7]. Failing to account for long-term sequences inhibits the deep learning model’s ability to generalize traffic patterns well and can limit prediction accuracy. Furthermore, most methods in the literature focus on 5-15 minute ahead speed prediction, which provides a short window for reaction to predicted traffic turbulence, and by extension, a limited practical advantage to potential users.

Secondly, most research efforts treat their deep learning models as “black box” models and thus fail to understand the reasons behind their model’s prediction processes and results [8], [9]. The lack of interpretability leads to a number of problems. For instance, it might discourage decision makers from deploying the model due to the absence of output explicability. Furthermore, the scarce interpretability impedes the model development, debugging and dissection of where performance deficiencies may be coming from.

In this paper, an attention-based multi-encoder-decoder model, dubbed Att-MED, is proposed to model and predict freeway traffic speed for up to a 60-minute horizon. The encoder component uses convolutions to model the spatial

dependency, and Long-Term-Short-Memory (LSTM) to model the temporal dependency. Additionally, Att-MED is capable of incorporating multiple encoders and is therefore able to extract traffic patterns from multiple input sequences with different periodicity. The decoder also uses an LSTM to model the output horizon sequentially. Furthermore, attention mechanism is used as an intermediate component between the encoder and decoder to enhance the temporal context of the decoder. Moreover, the attention weights are visualized to help decipher the neural network's prediction process. The proposed model is trained and tested on traffic data from January 1st, 2017 to June 30th, 2017 extracted from State Road 408 in Orlando, Florida, USA. In summary, the contribution of this paper is threefold:

- 1) The methodology demonstrates an attention-based multi-encoder-decoder model (Att-MED) that has the ability to encode multiple input traffic sequences. This allows the model to capture the different periodic characteristics of traffic flow, namely short-term, daily and weekly characteristics.
- 2) The Att-MED model was trained and tested for up to 60-minute-ahead network-wide speed prediction. The model performance was compared against baseline models. Moreover, the contribution of the attention layer towards prediction accuracy was quantified by implementing a version of the network without attention.
- 3) The paper enhances the interpretability of the proposed model by leveraging attention mechanisms to extract and visualize temporal dependencies in speed prediction. The visualization provides a deeper understanding of how the network operates and allows for nuanced and meticulous conclusions to be drawn from the results.

II. LITERATURE REVIEW

In the past decades, a number of parametric models have been used to predict traffic speed. Several works have implemented Autoregressive Integrated Moving Average Model (ARIMA) for time series prediction [10–12]. Additionally, multiple variations of ARIMA were implemented. For example, ARIMA with explanatory variables (ARIMAX) [13], seasonal ARIMA (SARIMA) [14], space-time ARIMA [15], and Kohonen-ARIMA (KARIMA) [16]. However, ARIMA models are not the best suited to capture the nonlinear spatiotemporal relationships of traffic speed since the model is built on linear regression concepts. Therefore, the implementations underutilize the learning data.

Conventional machine learning techniques have also been employed to tackle the speed prediction problem. For instance, linear conditional Gaussian Bayesian Networks [17], Support Vector Regression (SVR) [18]–[20], k-Nearest Neighbors (k-NNs) [21], [22] and ANNs [8], [23]–[25]. Castro-Neto *et al.* [18] proposed an SVR model to predict short-term traffic flow in typical and atypical traffic conditions and found that the SVR model performed better compared to ANNs. Davis and Nihan [22] used k-NNs to forecast freeway traffic and found that they produced comparable results to time-series linear regression models like ARIMA. Ye *et al.* [23] used GPS data to extract speed and acceleration

data on a road segment during irregular intervals. The data was then fed to an ANN that was used to forecast short-term traffic speed (60 seconds ahead). Deeper neural networks have also been applied to the traffic speed prediction problem. Lv *et al.* [26] proposed a Stacked Autoencoder (SAE) architecture to model the spatial and temporal traffic flow patterns. Their results outperformed shallow Backpropagation Neural Networks (BP-NNs) and SVR.

Recurrent Neural Networks (RNNs) are a class of deep learning algorithms that were designed to model sequential input and can possibly be utilized to construct sequential outputs. In the recent past, RNNs have been widely used to forecast traffic speed [25], [27]–[32], especially popular variations like Long Short Term Memory (LSTM) [33] and Gated Recurrent Units (GRUs) [34]. These models gained popularity due to their ability to memorize long-term sequential dependencies. Ma *et al.* [28] was one of the first to use LSTMs for short-term traffic speed prediction (up to 8 minutes). The LSTM outperformed ARIMA, SVM and non-recurrent variations of neural networks. Zhao *et al.* [31] modeled the temporal component using LSTMs and the special component using a correlation matrix between detector locations. Fu *et al.* [27] compared the performance of LSTMs and GRUs and reached slightly better results using GRUs. Cui *et al.* [32] used stacked bidirectional and unidirectional LSTMs (SBU-LSTMs) to perform network-wide speed prediction. Bidirectional recurrent units are able to capture temporal dependencies in both chronological and reverse chronological order.

Convolutional Neural Networks (CNNs) have been exceptionally successful at executing computer vision and image processing tasks in the past decade [35]. They have since been leveraged to capture the spatial dependencies across the road network in speed forecasting tasks [6], [7], [36]–[41]. Ke *et al.* [37] used CNNs to capture the special correlation between traffic in adjacent lanes and subsequently predict lane-wise traffic flow. Cao *et al.* [7] extracted the traffic spatial features from the road network using CNNs and then used the derived features as input for an LSTM, which in turn captured the temporal correlation. Liu *et al.* [6] combined convolutions and LSTMs as a single Conv-LSTM neural network architecture (first proposed by Xingjian *et al.* [42]) that is capable of both learning spatial and temporal dependencies. Furthermore, spatial dependencies were extracted using graph convolution networks (GCN) [9], [43]–[45]. Graph convolutions rely on extracting spatial features by convolving the directed graph representation of the road network. In this representation, detectors are represented by nodes and adjacent detectors are connected by edges. Spatio-Temporal Graph Convolutional Network (STGCN) was proposed by Yu *et al.* [44] to extract both spatial and temporal traffic features and subsequently using those features to predict traffic flow by applying graph convolution.

Attention mechanisms were first introduced in the natural language processing field and gained popularity as a result of their ability to determine and preserve sequence-wide dependencies [46], [47]. Previous works employed attention mechanism for traffic prediction [9], [45], [48]. This paper extends the research done on attention-based traffic prediction

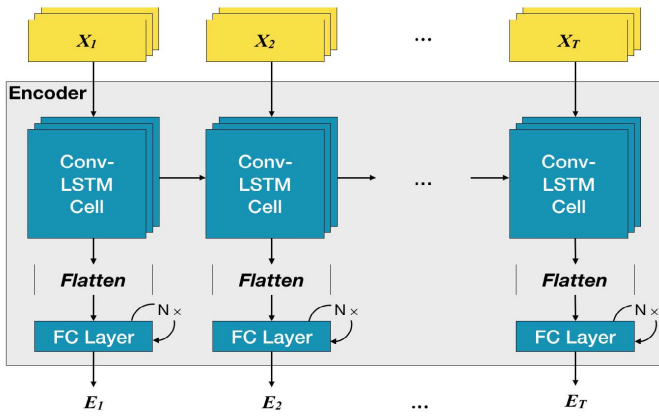


Fig. 1. Encoder neural network architecture.

by visualizing and interpreting the temporal dependencies of the short-term, daily and weekly input sequences captured by the attention weights.

III. METHODOLOGY

The network proposed by the research in this paper tackles the problem of speed prediction by utilizing 3 input sequences. First, a short-term sequence of length $T(\mathcal{X}_{t-T+1}, \dots, \mathcal{X}_{t-1}, \mathcal{X}_t)$, which captures the traffic features from the previous timesteps. Next, a daily sequence of length $D(\mathcal{X}_{d-D}, \dots, \mathcal{X}_{d-2}, \mathcal{X}_{d-1})$, which lists the traffic features during the same time-of-day for the past D days. Finally, a weekly sequence of length $W(\mathcal{X}_{w-W}, \dots, \mathcal{X}_{w-2}, \mathcal{X}_{w-1})$, which captures the traffic features during the same time-of-day in the same day-of-week for the past W weeks. Each element $\mathcal{X}$ in the input sequence contains the traffic features of all road segments in the freeway network. The model forecasts traffic speed for the next M timesteps $(\mathcal{Y}_{t+1}, \mathcal{Y}_{t+2}, \dots, \mathcal{Y}_{t+M})$. Each element $\mathcal{Y}$ in the output sequence contains the network-wide traffic speed predictions (i.e., all locations in the freeway network). The proposed network consists of 3 main components: encoder, attention layer and decoder.

A. Encoder

The encoder combines 2 neural network algorithms: convolution is used to extract the spatial features between traffic detectors and LSTM is used to model the temporal dependency. The two modules are combined into one network architecture called Conv-LSTM. A Conv-LSTM accepts a 4-dimensional input of the following structure: (time sequence, features, width, length). Time sequence is the temporal input sequence dimension. The feature vector contains the traffic parameters pertaining to each specific location in the network at a given time in the sequence. Finally, the width and length dimensions capture the spatial characteristics of the freeway network. To model the last 2 dimensions, the traffic parameters at each timestep have been reshaped into a vector of size (width, length) where width = 1 and length = number of road segments in the freeway network.

The convolution module focuses on extracting the spatial features from the road network and hence computes the

features from each time slice separately. The module accepts a 3D tensor in $\mathbb{R}^{F \times W \times L}$ as input where the tensor represents features, width, and length. The temporal feature extraction, handled by the LSTM module, involves computing cell outputs $C_1 \dots C_T$ and hidden states $\mathcal{H}_1 \dots \mathcal{H}_T$ for each 3D tensor in the input sequence $\mathcal{X}_1 \dots \mathcal{X}_T$. At each time step, the LSTM gates i_t, f_t and o_t decide how much information should be carried over from the previous cell output C_{t-1} and hidden state $\mathcal{H}_{t-1}$ and how much information can be drawn from the current input $\mathcal{X}_t$. Weight matrices W and bias vectors b are the Conv-LSTM trainable parameters, the values of which are determined by the backpropagation algorithm [49]. In summary, the Conv-LSTM is governed by the following set of equations, where $\sigma(\cdot)$ is the sigmoid function, $*$ denotes a convolution operation, and $\diamond$ computes the Hadamard product (also known as the element-wise product).

$$i_t = \sigma(W_{xi} * \mathcal{X}_t + W_{hi} * \mathcal{H}_{t-1} + W_{ci} \diamond C_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(W_{xf} * \mathcal{X}_t + W_{hf} * \mathcal{H}_{t-1} + W_{cf} \diamond C_{t-1} + b_f) \quad (2)$$

$$C_t = f_t \diamond C_{t-1} + i_t \diamond \tanh(W_{xc} * \mathcal{X}_t + W_{hc} \diamond \mathcal{H}_{t-1} + b_c) \quad (3)$$

$$o_t = \sigma(W_{xo} * \mathcal{X}_t + W_{ho} * \mathcal{H}_{t-1} + W_{co} \diamond C_t + b_o) \quad (4)$$

$$\mathcal{H}_t = o_t \diamond \tanh(C_t) \quad (5)$$

To decrease the dimensionality of the Convolutional-LSTM output at each time step, the output is flattened and then passed to a number of fully connected (FC) layers with ReLU activation. The encoder network produces a sequential output $E_1 \dots E_T$ as shown in the equations below. The full, unrolled encoder network structure is demonstrated in Fig. 1.

$$FC(1) = \text{ReLU}(W_{1t} \mathcal{H}_t + b_{1t}) \quad (6)$$

$$FC(N) = \text{ReLU}(W_{Nt} FC(N-1) + b_{Nt}) \quad (7)$$

$$E_t = FC(N) \quad (8)$$

B. Attention Layer

The attention mechanism enhances the encoder-decoder network by allowing each step of the decoder to assign attention weights to the input sequence. The more relevant an encoder step in the sequence is, the higher the weight of the assigned attention. The proposed network utilizes the attention mechanism to compute the relevance of short-term, daily and weekly periodicity when predicting each output horizon in the output sequence.

To compute the decoder input at timestep t , the attention energy of the encoder output at timestep i , denoted as e_{ti} , is calculated as a function of the previous decoder hidden state S_{t-1} and the output of the encoder at timestep i , denoted as E_i . The attention energies of all the encoder outputs are calculated and subsequently passed through a softmax function to create attention weights α_{ti} . The softmax function ensures that the aggregate of the assigned attention weights sums to 1. Lastly, a single vector is calculated for the output attention at time t . It is computed as the dot product of the attention weights and the encoder outputs. The final attention layer output, called the context vector, is passed as input to the decoder layer. In essence, the attention weights are computed according to the equations below, where weight matrices W

and bias vector b are the attention layer trainable parameters. The $\cdot$ operator calculates matrix multiplication and the $\exp(\cdot)$ function computes the exponent.

$$e_{ti} = W_{ti} \cdot \tanh(W_{st} \cdot \mathcal{S}_{t-1} + W_{ei} \cdot \mathcal{E}_i) + b_{ti} \quad (9)$$

$$a_{ti} = \frac{\exp(e_{ti})}{\sum_{k=1}^K \exp(e_{tk})} \quad (10)$$

$$CXT_t = \sum_{i=1}^I a_{ti} \mathcal{E}_{ti} \quad (11)$$

C. Decoder

The decoder layer consists of an LSTM neural network which computes the output sequence $\mathcal{Y}_1 \dots \mathcal{Y}_T$. The network uses the context vector CXT_t as input at each output timestep t . The LSTM computes its internal hidden state $\mathcal{S}_t$ and sends it to the attention unit of the next timestep. The i_t , f_t and o_t LSTM gates decide how much information to use from the context vector and how much to use from the previous decoder time steps. The output of each decoder cell $\mathcal{Y}_t$ is a vector in $\mathbb{R}^N$ where N is the number of detectors on the road network. This enables the LSTM to predict network-wide traffic speed at each output timestep.

$$i_t = \sigma(W_{ix} \cdot CXT_t + W_{is} \cdot \mathcal{S}_{t-1} + W_{ic} \cdot c_{t-1} + b_i) \quad (12)$$

$$f_t = \sigma(W_{fx} \cdot CXT_t + W_{fs} \cdot \mathcal{S}_{t-1} + W_{fc} \cdot c_{t-1} + b_f) \quad (13)$$

$$c_t = f_t \diamond c_{t-1} + i_t \diamond \tanh(W_{cx} \cdot CXT_t + W_{cs} \cdot \mathcal{S}_{t-1} + b_c) \quad (14)$$

$$o_t = \sigma(W_{ox} \cdot CXT_t + W_{os} \cdot \mathcal{S}_{t-1} + W_{oc} \cdot c_t + b_o) \quad (15)$$

$$\mathcal{S}_t = o_t \diamond \tanh(c_t) \quad (16)$$

$$\mathcal{Y}_t = W_{ys} \cdot \mathcal{S}_t + b_y \quad (17)$$

To summarize, the decoder computes the output sequence according to the above set of equations. As mentioned above, $\sigma(\cdot)$ is the sigmoid function, $\cdot$ denotes matrix multiplication and $\diamond$ calculates the Hadamard product. Matrices W and vectors b are the decoder trainable parameters.

D. Att-MED Network

To model the multi-input-sequence to output-sequence problem, the Att-MED model constructs 3 encoders: short-term, daily and weekly. Each encoder extracts the spatiotemporal parameters of its assigned input sequence. Next, the encoder outputs are concatenated and subsequently used as an input to the attention layer. The attention layer weighs the importance of each value in each input sequence and computes a combined context vector, which is then passed as an input to the decoder network. Each decoder LSTM cell forwards its hidden state as an input to the next attention unit. Fig. 2 illustrates how the network computes the network-wide traffic speed as an output sequence by encoding multiple input sequences.

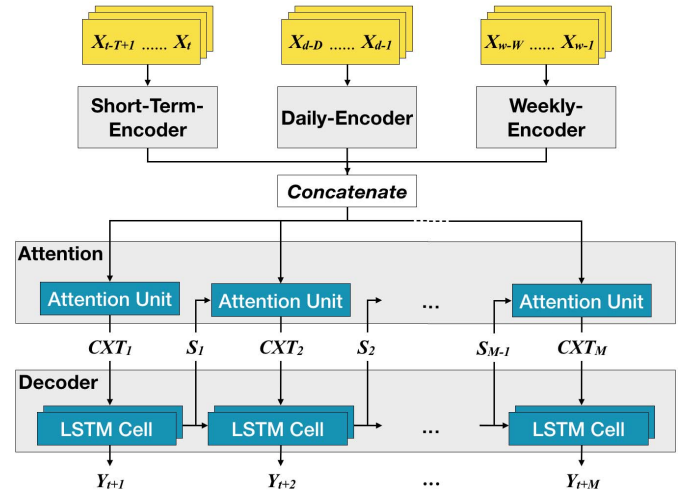


Fig. 2. Att-MED network architecture.

TABLE I
INPUT VARIABLES SUMMARY STATISTICS

	Min	Max	Mean	Standard Deviation
Speed (mph)	0.00	149.08	65.35	6.37
Volume	0.00	416.00	141.59	116.23

the study focuses on the east-bound direction. There are 52 Microwave Vehicle Detection System (MVDS) sensors in the selected area, which are spaced 0.4 miles apart on average. The total length of the covered route is 20.8 miles. Fig. 3 depicts their distribution along the freeway. The microwave sensors record spot speed, total volume and average occupancy every 60 seconds. The MVDS historical data has been archived on the Regional Integrated Transportation Information System (RITIS) and has been downloaded for this paper from the RITIS website.

The time range of the historical data used for this study is January 1st, 2017 to June 30th, 2017. To take advantage of the daily traffic patterns, the study focuses on speed prediction during weekdays. Based on the Speed-Density-Flow traffic relationship, 2 of the detected traffic parameters can be used to compute the third [50]. Therefore, to diminish the potential irreducible error that may be caused by detector inaccuracies, only speed and volume were chosen as traffic features. Speed and volume data were aggregated to 5-minute intervals. The summary statistics of the per-segment aggregated features are presented in Table I. The collected traffic dataset was split into a training set, a validation set, and a testing set into a ratio of 50%:25%:25%, respectively. The validation set was utilized in the deep learning models.

B. Performance Metrics

The experiment used three widely applied evaluation metrics to qualify the performance of each model. They are Mean Absolute Error (MAE), Root Mean Square Error (RMSE) and

IV. EXPERIMENTATION

A. Data Preparation

The experiment studied traffic speed along State Road 408, an east-west freeway in Orlando, Florida, USA. Furthermore,

TABLE II
SHORT-TERM SPEED PREDICTION RESULTS

Model	5-minutes			10-minutes			15-minutes		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
ARIMA	1.580	2.946	2.721	1.716	3.419	3.030	1.953	3.814	3.315
SVR	2.610	5.369	4.075	2.609	5.262	4.077	2.613	5.269	4.081
ANN	1.634	2.927	2.794	1.847	3.420	3.153	1.871	3.635	3.220
Bi-LSTM	1.810	3.730	3.023	1.852	3.750	3.344	1.959	3.867	3.380
Conv-LSTM	1.605	2.979	2.656	1.699	3.181	2.835	1.865	3.398	2.998
Att-MED	1.451	2.553	2.240	1.476	2.617	2.320	1.539	2.674	2.404

Mean Absolute Percentage Error (MAPE). In the following equations, $\hat{y}_i$ refers to a prediction made by the model while y_i refers to its corresponding ground-truth value. The total number of data points used for evaluation is denoted as n .

$$MAE = \frac{1}{n} \sum_{i=0}^n |y_i - \hat{y}_i| \quad (18)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=0}^n (y_i - \hat{y}_i)^2} \quad (19)$$

$$MAPE = \frac{100}{n} \sum_{i=0}^n \frac{|y_i - \hat{y}_i|}{y_i} \quad (20)$$

C. Baseline Models

The experiment target was to, at a given time t , predict the speed values of the next time increments up to 60 minutes. Since the processed data has been resampled as 5-minute intervals, the objective equated to predicting the speed values of the next 12 timesteps. Moreover, each MVDS detector represented a road segment in the experiment. The results of our model are compared against the following models:

- **ARIMA**: Auto-regressive integrated moving average.
- **SVR**: Support Vector Regression using a radial basis function (RBF) kernel.
- **ANN**: A deep, feed forward neural network. The hyperparameters are the number of layers and the number of units per layer.
- **Bi-LSTM**: An encoder-decoder Bidirectional LSTM network. The hyperparameters are the number of units in the encoder and decoder layers.
- **Conv-LSTM**: A sequence-to-sequence model which uses a convolutional LSTM network as an encoder and an LSTM as a decoder. The number of Conv-LSTM units, kernel size, as well as the number of LSTM units are the model hyperparameters.

As for the Att-MED model, the number of units in the Conv-LSTM encoder cells, LSTM decoder cells as well as the number of encoder FC layers are considered hyperparameters. To include both traffic speed and volume in the

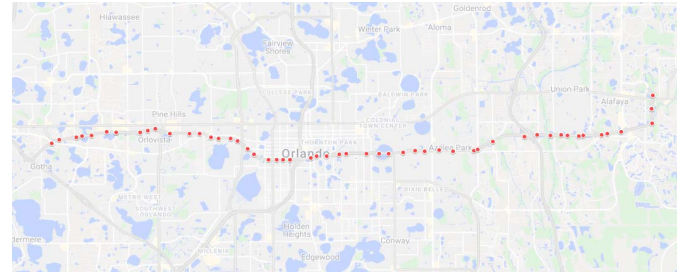


Fig. 3. The distribution of detectors along the freeway.

input space, and to accommodate the shape of the freeway network displayed in Fig. 3, the features, width, and length input parameters were assigned sizes 2, 1, and 52, respectively. Furthermore, to enable network-wide speed detection, the size of the model's output at each prediction timestep was set to 52.

All neural network models were implemented using Tensorflow 2.2.1 [51]. The hyperparameters were tuned using the random search implementation of the Keras-Tuner package [52]. The tuner used validation loss as a comparison metric to avoid optimizing the hyperparameters to overfit the training data. Furthermore, the Adam optimizer [53] was employed, and the loss function was set to MAE. The models ran for 250 epochs. Early stopping was utilized to prevent overfitting by monitoring the validation loss. It prevents overtraining by monitoring the trend of the validation loss. If the validation loss deteriorates for a certain number of epochs, the training process is halted even if the training loss continues to decrease.

D. Results

Tables II and III shows the prediction results of the proposed model and baseline models on the testing dataset for the 52 road segments. Table II displays the short-term results (5min/10min/15min), while Table III focuses on longer term predictions (30min/45min/60min). The results indicate that the SVR model was not able to model the temporal dependency correctly. The ARIMA model was able to produce good short-term results, but the performance quickly deteriorated when predicting farther time horizons, indicating that it's unable to handle long-term temporal dependencies. The Att-MED,

TABLE III
LONG-TERM SPEED PREDICTION RESULTS

Model	30-minutes			45-minutes			60-minutes		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE	MAE	RMSE	MAPE
ARIMA	2.157	4.752	4.080	2.433	5.486	4.752	2.689	6.106	5.375
SVR	2.617	5.281	4.088	2.630	5.305	4.108	2.649	5.345	4.138
ANN	2.075	4.028	3.602	2.140	4.363	3.752	2.295	4.770	5.450
Bi-LSTM	2.044	4.210	3.567	2.154	4.472	3.751	2.254	4.684	3.949
Conv-LSTM	1.976	3.885	3.353	2.067	4.091	3.542	2.120	4.180	3.643
Att-MED	1.582	2.801	2.548	1.671	3.034	2.703	1.777	3.376	2.899

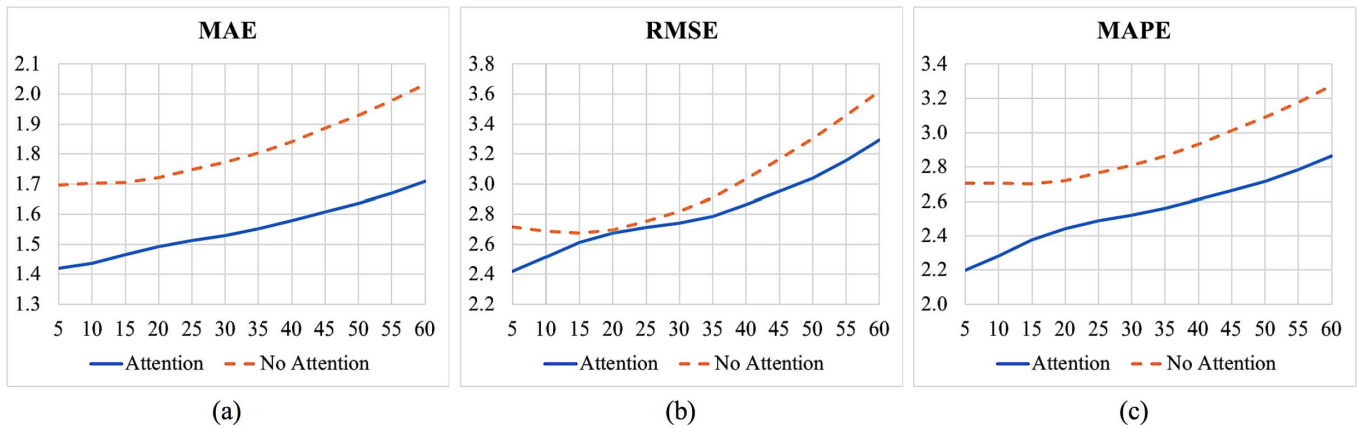


Fig. 4. Prediction results for the proposed model with and without attention (a) MAE (b) RMSE (c) MAPE.

Bi-LSTM and Conv-LSTM models performed better than the basic ANN, highlighting the importance of capturing the temporal dependency provided by the recurrent neural networks structure. Furthermore, the proposed model and the Conv-LSTM's convolution modules were able to model the spatial dependencies in the road network.

Att-MED generated superior results to all the baseline models for every prediction horizon. This outcome illustrates the significance of the daily and weekly traffic parameter periodicity as input in addition to the short-term periodicity. Furthermore, attention is needed to regulate the weight of each input sequence during prediction. Fig. 4 depicts the results of the proposed model with and without the attention layer. The results indicate that the attention mechanism balanced between the input sequences to minimize the prediction error.

E. Attention Visualization

Neural network models are often criticized for operating as “black boxes” and lacking interpretability. To increase model explainability, the attention weights assigned to prediction samples were extracted and visualized. Figs. 5a, 5b and 5c map the attention weights of the Att-MED model when making a prediction of a morning peak sample (9:00 am), off-peak sample (1:00 pm) and nighttime sample (3:00 am). In order

to avoid the potential training bias associated with the training and validation sets, the visualized attention weights were calculated for input data sampled from the testing set. The chosen values for T , D and W were 12, 5 and 2, respectively. Since attention weights pass through a softmax function, the aggregate value of each column in the plots is 1. It can be determined that the model relied on weekly periodicity while making predictions during peak hour. The model relied more on both daily and weekly cyclic characteristics to forecast the speed of off-peak samples. In contrast, the model shifted more focus toward short-term traffic features to make predictions during nighttime.

Fig. 6 illustrates a sample of daily and weekly traffic speed patterns on State Road 408 milepost 09.6. Fig. 6a indicates that, during the morning peak, Friday and Thursday patterns differ slightly from the rest of the weekdays. On the other hand, weekly peak traffic rhythm seems to be closer as demonstrated by Fig. 6b. This pattern validates the model's choice to focus on weekly traffic patterns when making predictions during peak hours. The same can be said for nighttime traffic, which exhibits high variability across both daily and weekly cycles. The model chose to focus the attention weights on short-term features due lack of strong cyclic traffic patterns during nighttime. As for off-peak hours, specifically between the morning and evening peak, Fig. 6 indicates that traffic speed exhibits a similarity across daily and weekly patterns.

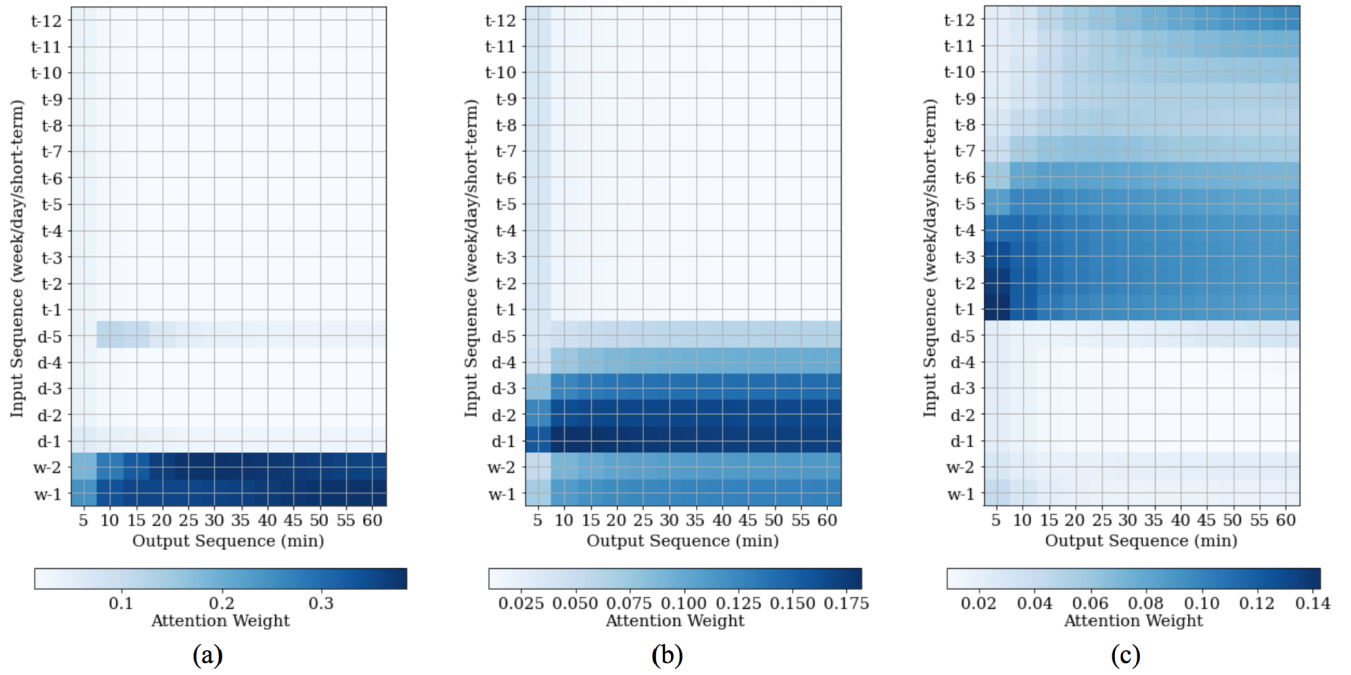


Fig. 5. Attention weights of proposed model predictions during (a) morning peak hour sample (9 am) (b) off-peak hour sample (1 pm) (c) nighttime sample (3 am).

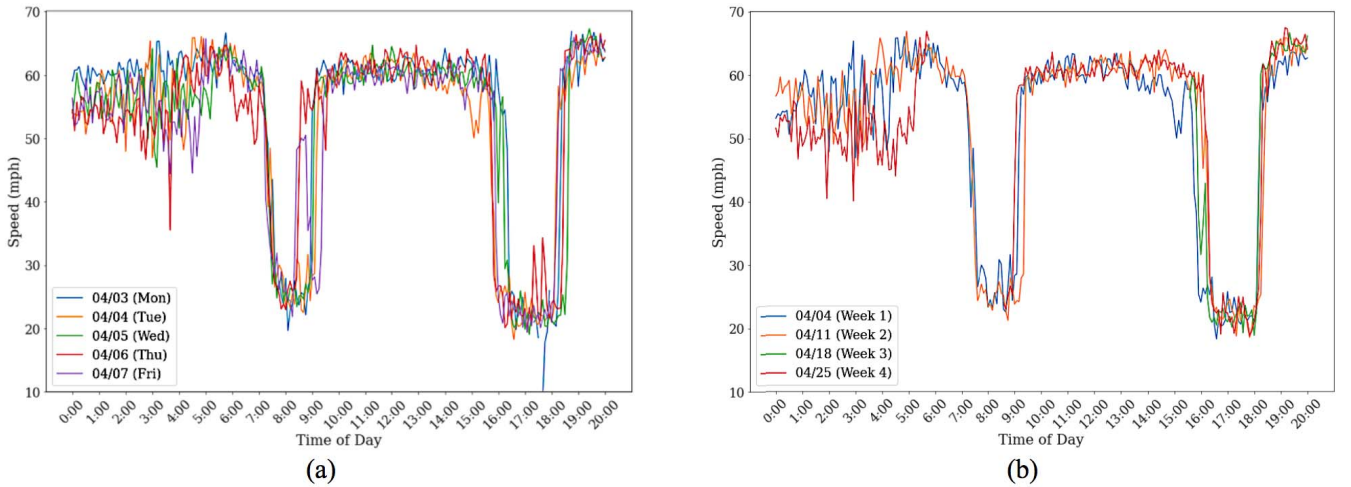


Fig. 6. (a) Daily and (b) weekly trends of traffic speed at freeway milepost 09.6.

Fig. 5b demonstrates that this similarity was captured by the attention weights.

Figs. 7a, 7b and 7c map the attention weights throughout a sample day for 5-minute, 30-minute and 60-minute attention horizons, respectively. Similar to Fig. 5, the data samples used for visualization were extracted from the testing set. The plots illustrate the model's choice to focus on weekly periodic characteristics during evening peak hours and a combination of daily and weekly cyclic features for off-peak hours. It can also be observed from Fig. 7, as well as Figs. 5a and 5b, that the model relied more on short-term features when forecasting speed for a shorter output horizon. As the model makes

farther predictions, it relies more on daily and weekly features. Shorter horizons are inherently easier to predict, given that there is less variability between the prediction and short-term features. Furthermore, assigning more weight to short-term features allows the model to react to atypical conditions like crashes, especially for shorter-horizon speed forecasting which would be impacted the most by sudden abnormalities in traffic conditions. On the contrary, longer-term traffic forecasting is inherently more stochastic. Given that there's more variability between short-term traffic features and longer prediction horizons, the attention layer assigned a heavier weight to daily and weekly patterns.

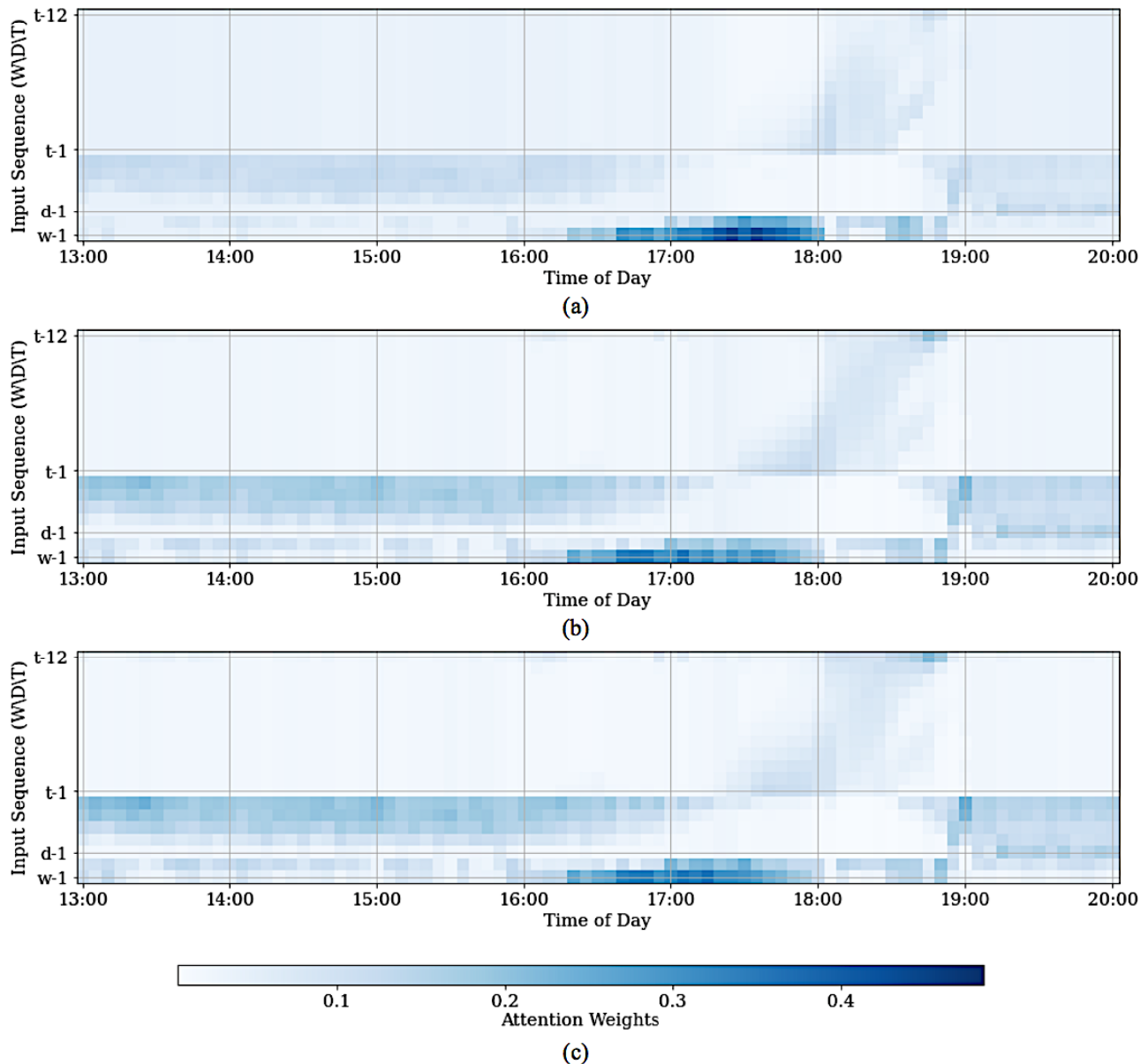


Fig. 7. Attention weights of the prediction of a sample time segment for (a) 5-minute (b) 30-minute and (c) 60-minute output horizons.

V. CONCLUSION

The architecture of the proposed Att-MED model is composed of multiple components. The encoder used convolutions and a recurrent neural network cells to capture the spatiotemporal relationship of the input sequence. The usage of multiple encoders helped capture the cyclic daily and weekly traffic patterns and facilitated their use as additional input features. The attention layer enabled the use of multiple encoders, and by extension, multiple traffic sequence inputs. Attention balanced between the different encoder outputs and regulated their contribution towards prediction at each output horizon. Finally, the LSTM decoder modeled the temporal relationship of the output sequence. Training the attention-based multi-encoder decoder network end-to-end produced superior 60-minute ahead prediction accuracies compared to the baseline models. Furthermore, the attention layer enhanced

the interpretability of the model outputs. The model prediction decisions were further studied and validated by comparing against the real traffic patterns. The attention weights confirmed the significance and contribution of the daily and weekly input sequences towards speed prediction. The attention weight maps quantified the value of periodic features, especially for longer-horizon prediction.

Accurate 60-minute ahead traffic speed prediction has many useful applications. For regular commuters and connected vehicles, it can be used for trip planning, routing and pre-emptive rerouting in case of predicted congestions. It can also be applied in traffic management, especially during peak hours. In practice, 60-minute ahead prediction gives traffic operators plenty of time to react to predicted traffic turbulence. Additionally, in contrast to traditional deep learning models, attention weight visualization enhances the interpretability of

the proposed model, which allows operators to make nuanced, informed, and explainable decisions.

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